

Java Loops Masterclass

50 Coding Exercises for Beginners

This document is designed to help beginner Java developers MASTER loops through repetition, tracing, debugging, and problem-solving. The goal is NOT to memorize syntax. The goal is to train your brain to think step-by-step like a programmer.

Important Rules for Students

- You MUST push every completed exercise to GitHub.
- Create a separate folder or file for each task.
- Do NOT use Artificial Intelligence tools to solve the exercises.
- Trace your code manually before running it.
- Write comments explaining your thinking.
- If your program fails, debug it before asking for help.
- Use meaningful variable names.
- Practice every day, even if only for 30 minutes.
- Do not copy from classmates.
- Focus on understanding WHY the loop works.

Level 1 — Understanding Basic Loops

#	Exercise
1	Print numbers from 1 to 10 using a for loop.
2	Print numbers from 10 down to 1.
3	Print all even numbers from 1 to 20.
4	Print all odd numbers from 1 to 20.
5	Print your name 15 times.
6	Calculate the sum of numbers from 1 to 100.
7	Calculate the product of numbers from 1 to 5.
8	Print all multiples of 3 between 1 and 50.
9	Print numbers divisible by both 2 and 5.

10	Print the square of numbers from 1 to 10.
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Level 2 — While Loops and Logic

#	Exercise
11	Use a while loop to print numbers from 1 to 20.
12	Use a while loop to print even numbers from 2 to 30.
13	Count backwards from 50 to 1.
14	Calculate the sum of even numbers from 1 to 50.
15	Ask the user for 5 numbers and calculate the total.
16	Keep asking the user for a password until they enter the correct one.
17	Create a countdown timer from 10 to 0.
18	Print the multiplication table of 7.
19	Find the largest number from 5 user inputs.
20	Find the smallest number from 5 user inputs.

Level 3 — Nested Loops

#	Exercise
21	Print a square pattern of stars.
22	Print a right triangle pattern.
23	Print an inverted triangle pattern.
24	Print a pyramid pattern.
25	Print a multiplication table from 1 to 10.
26	Print coordinates using nested loops.
27	Print a checkerboard pattern using X and O.
28	Print numbers in rows and columns.
29	Print a rectangle using stars.
30	Print a hollow square pattern.

Level 4 — Loop Thinking and Problem Solving

#	Exercise
31	Find the factorial of a number.
32	Reverse a number using loops.

33	Check whether a number is prime.
34	Count the digits in a number.
35	Find the sum of digits in a number.
36	Generate the Fibonacci sequence.
37	Check whether a number is a palindrome.
38	Find the greatest common divisor of two numbers.
39	Print all prime numbers from 1 to 100.
40	Find the average of user-entered numbers.

Level 5 — Advanced Beginner Challenges

#	Exercise
41	Build a simple ATM menu system using loops.
42	Create a number guessing game.
43	Simulate a login system with 3 attempts.
44	Create a basic calculator that runs continuously until the user exits.
45	Create a menu-driven restaurant ordering system.
46	Create a grading system using loops and switch.
47	Build a student attendance tracker.
48	Generate a bar chart using stars.
49	Print a diamond star pattern.
50	Create a mini banking simulation.

How to Actually Master Loops

- Trace every loop iteration on paper.
- Always ask: What changes after every iteration?
- Understand the condition that stops the loop.
- Practice nested loops slowly.
- Predict the output BEFORE running the code.
- Break large problems into smaller loop problems.
- Learn from errors instead of fearing them.
- Consistency beats intensity.

The students who become excellent engineers are usually not the smartest people in the room. They are the people who practiced fundamentals repeatedly until they became second nature. Loops are one of those fundamentals.